

Section 1.1 Exercises

For the exercises 1 - 4, evaluate f at the indicated values $f(-3)$, $f(2)$, $f(-a)$, $-f(a)$, $f(a + h)$.

1. $f(x) = 2x - 5$

2. $f(x) = -5x^2 + 2x - 1$

3. $f(x) = \sqrt{2 - x} + 5$

4. $f(x) = \frac{6x-1}{5x+2}$

5. Given the function $g(x) = 5 - x^2$, evaluate $\frac{g(x+h)-g(x)}{h}$, $h \neq 0$.

6. Given the function $g(x) = x^2 + 2x$, evaluate $\frac{g(x)-g(a)}{x-a}$, $x \neq a$.

7. Given the function $k(t) = 2t - 1$:

a. Evaluate $k(2)$.

b. Solve $k(t) = 7$.

8. Given the function $f(x) = 8 - 3x$:

a. Evaluate $f(-2)$.

b. Solve $f(x) = -1$.

9. Given the function $p(c) = c^2 + c$:

a. Evaluate $p(-3)$.

b. Solve $p(c) = 2$.

10. Given the function $f(x) = x^2 - 3x$:

a. Evaluate $f(5)$.

b. Solve $f(x) = 4$.

11. Given the function $f(x) = \sqrt{x + 2}$:

a. Evaluate $f(7)$.

b. Solve $f(x) = 4$.

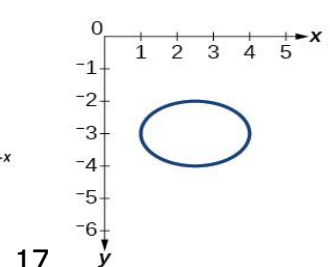
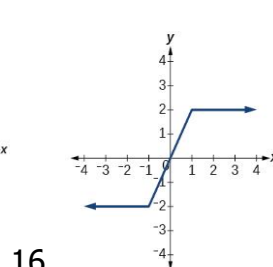
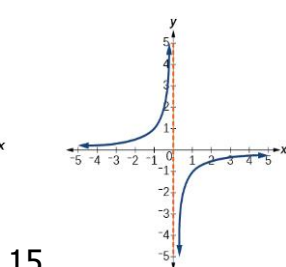
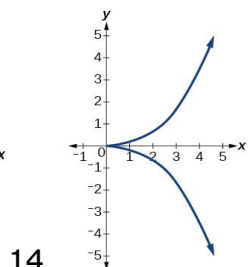
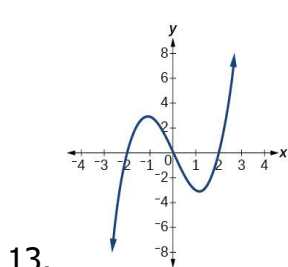
12. Consider the relationship $3r + 2t = 18$.

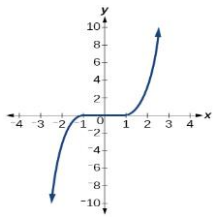
a. Write the relationship as a function $r = f(t)$.

b. Evaluate $f(-3)$.

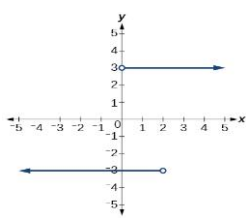
c. Solve $f(t) = 2$.

For the exercises 13 - 22, use the vertical line test to determine which graphs show relations that are functions.

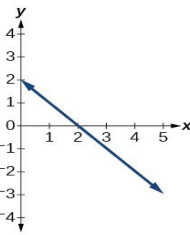




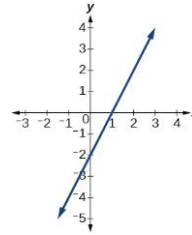
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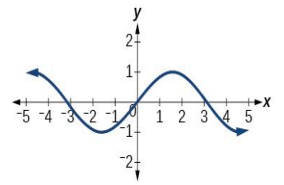
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20.



21.

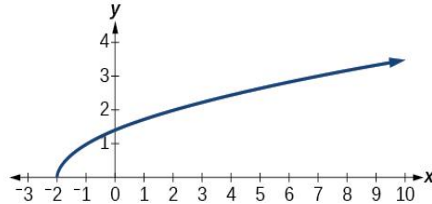


22.

23. Given the graph,

a. Evaluate $f(-1)$.

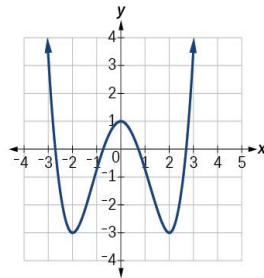
b. Solve for $f(x) = 3$.



24. Given the graph,

a. Evaluate $f(0)$.

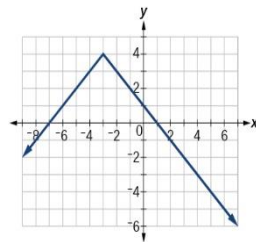
b. Solve for $f(x) = -3$.



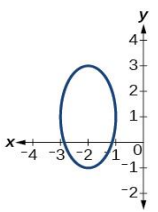
25. Given the graph,

a. Evaluate $f(4)$.

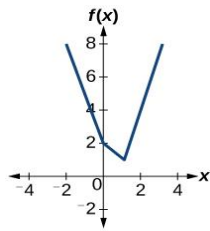
b. Solve for $f(x) = 1$.



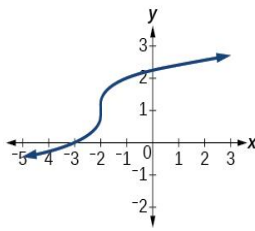
For the exercises 26 - 30, determine if the given graph is a one-to-one function.



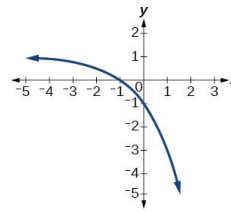
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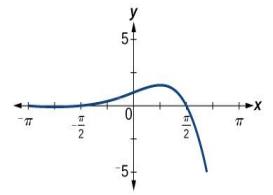
27.



28.



29.



30.

For the exercises 31 - 33, determine whether the relation represents a function.

31. $\{(-1, -1), (-2, -2), (-3, -3)\}$

32. $\{(3, 4), (4, 5), (5, 6)\}$

33. $\{(2, 5), (7, 11), (15, 8), (7, 9)\}$

For the exercises 34 - 36, determine if the relation represented in table form represents y as a function of x .

34.

x	5	10	15
y	3	8	14

35.

x	5	10	15
y	3	8	8

36.

x	5	10	10
y	3	8	14

For the exercises 37 - 38, use the function f represented in the following table.

x	0	1	2	3	4	5	6	7	8	9
$f(x)$	74	28	1	53	56	3	36	45	14	47

37. Evaluate $f(3)$.

38. Solve $f(x) = 1$.

For the exercises 39 - 44, evaluate the function f at the values $f(-2)$, $f(-1)$, $f(0)$, $f(1)$, and $f(2)$.

39. $f(x) = 4 - 2x$

40. $f(x) = 8 - 3x$

41. $f(x) = 8x^2 - 7x + 3$

42. $f(x) = 3 + \sqrt{x+3}$

43. $f(x) = \frac{x-2}{x+3}$

44. $f(x) = 3^x$

For the exercises 45 - 46, evaluate the expressions, given functions f , g , and h :

- $f(x) = 3x - 2$
- $g(x) = 5 - x^2$
- $h(x) = -2x^2 + 3x - 1$

45. $3f(1) - 4g(-2)$

46. $f\left(\frac{7}{3}\right) - h(-2)$